Lecture 1

Chapter 1

INTRODUCTION

A Few Historical Images

FIGURE 1.1 A digital picture produced in 1921 from a coded tape by a telegraph printer with special typefaces. (McFarlane.) [References in the bibliography at the end of the book are listed in alphabetical order by authors' last names.]

FIGURE 1.2 A digital picture made in 1922 from a tape punched after the signals had crossed the Atlantic twice. (McFarlane.)

A Few Historical Images

FIGURE 1.3 Unretouched cable picture of Generals Pershing (right) and Foch, transmitted in 1929 from London to New York by 15-tone equipment. (McFarlane.)

A Few Historical Images

FIGURE 1.4 The first picture of the moon by a U.S. spacecraft. *Ranger 7* took this image on July 31, 1964 at 9:09 A.M. EDT, about 17 minutes before impacting the lunar surface. (Courtesy of NASA.)

Electromagnetic Spectrum (Energy Source)

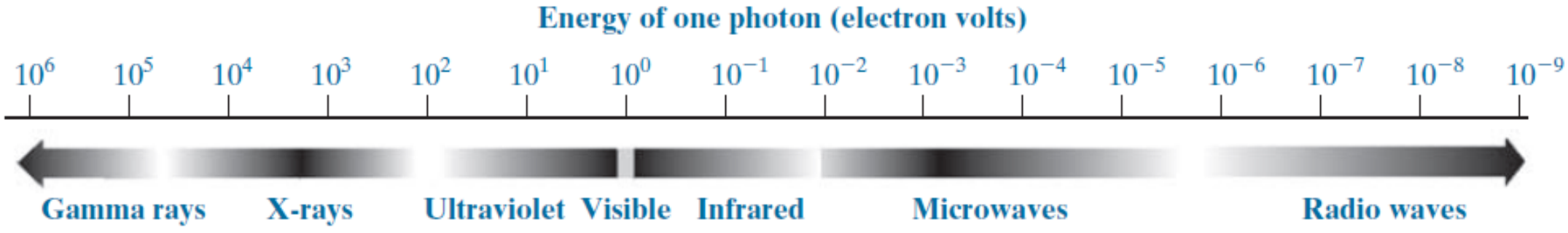

FIGURE 1.5 The electromagnetic spectrum arranged according to energy per photon.

- Human being can see only visual band of EM spectrum
- Images can be generated from the entire EM spectrum, ranging from Gamma to Radio waves

Gamma Ray Imaging

- Imaging based on gamma rays: Nuclear medicine injected into the patient with a radioactive isotope emits gamma rays. Images are produced from the emissions collected by gamma ray detectors.

- Positron Emission Tomography (PET): the patient is given a radioactive isotope that emits positron. Positron meets electron and two gamma rays are emitted which are detected by gamma ray detector.

- Constellation Sygnus: star exploded and produced a superheated gas cloud emitting energy in the entire EM spectrum

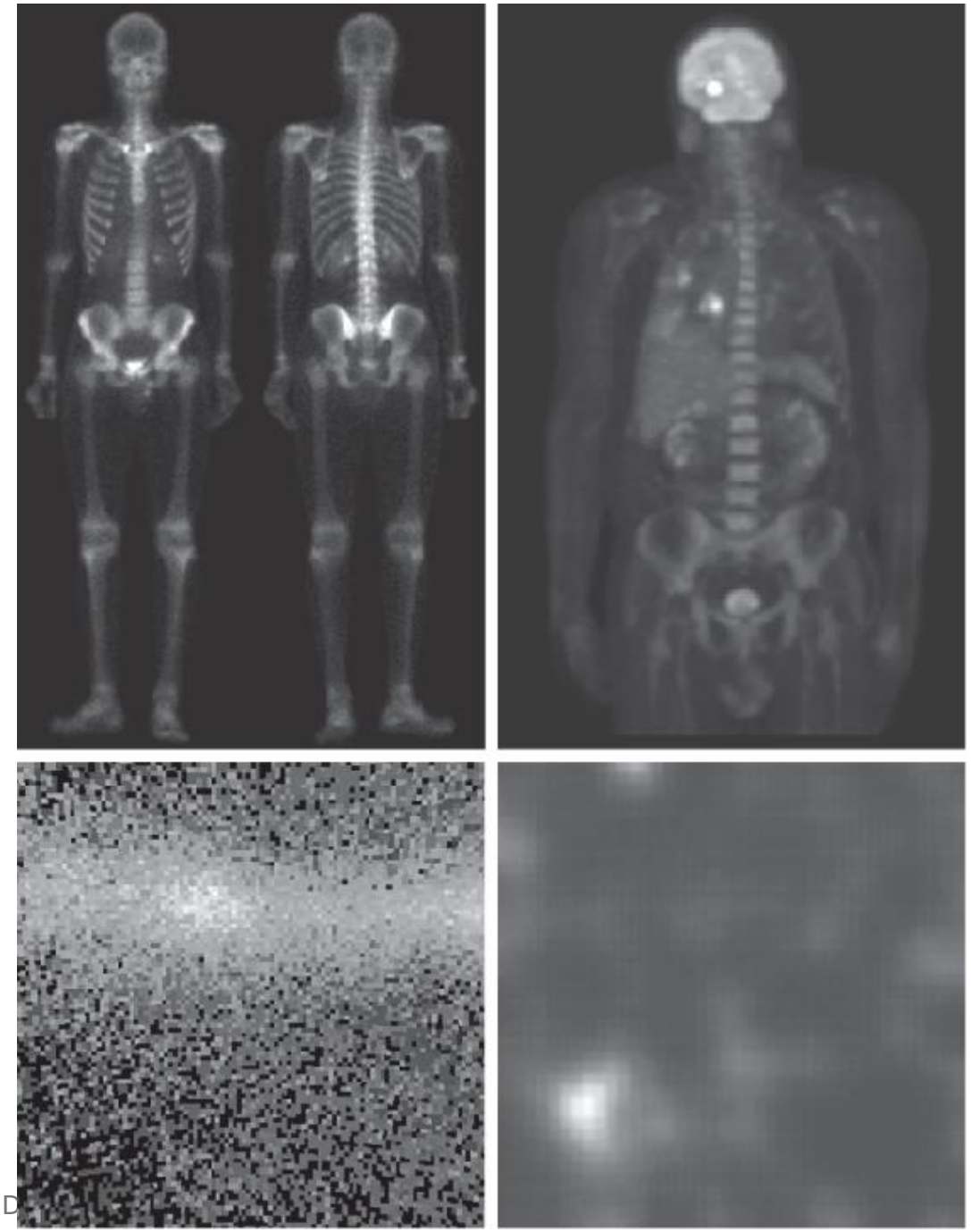

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FIGURE 1.6 Examples of gamma-ray imaging. (a) Bone scan. (b) PET image. (c) Cygnus Loop. (d) Gamma radiation (bright spot) from a reactor valve. (Images courtesy of (a) G.E. Medical Systems; (b) Dr. Michael E. Casey, CTI PET Systems; (c) NASA; (d) Professors Zhong He and David K. Wehe, University of Michigan.)

X-Ray Imaging

- X-ray tube: a cathode which is heated and releases electrons. Electrons fly at high speed to the positively charged anode. When the electrons strike a nucleus, energy is released in the form of X-ray radiation. The intensity of the X-ray is modified by absorption as it passes through the patient.
- Angiography: To obtain the images of blood vessels (Angiograms); used in Contrast-Enhancement Radiography. Catheter (a small, flexible, hollow tube) is inserted into artery/vein. When reached into area to be observed, a high contrast medium is inserted to enhance contrast to examine any irregularity/blockage.
- Computerized Tomography (CT) is an x-ray imaging procedure in which a narrow beam of x-rays is aimed at a patient and quickly rotated around the body, producing signals that are processed by the machine's computer to generate cross-sectional images, or "slices." These slices are called tomographic images.

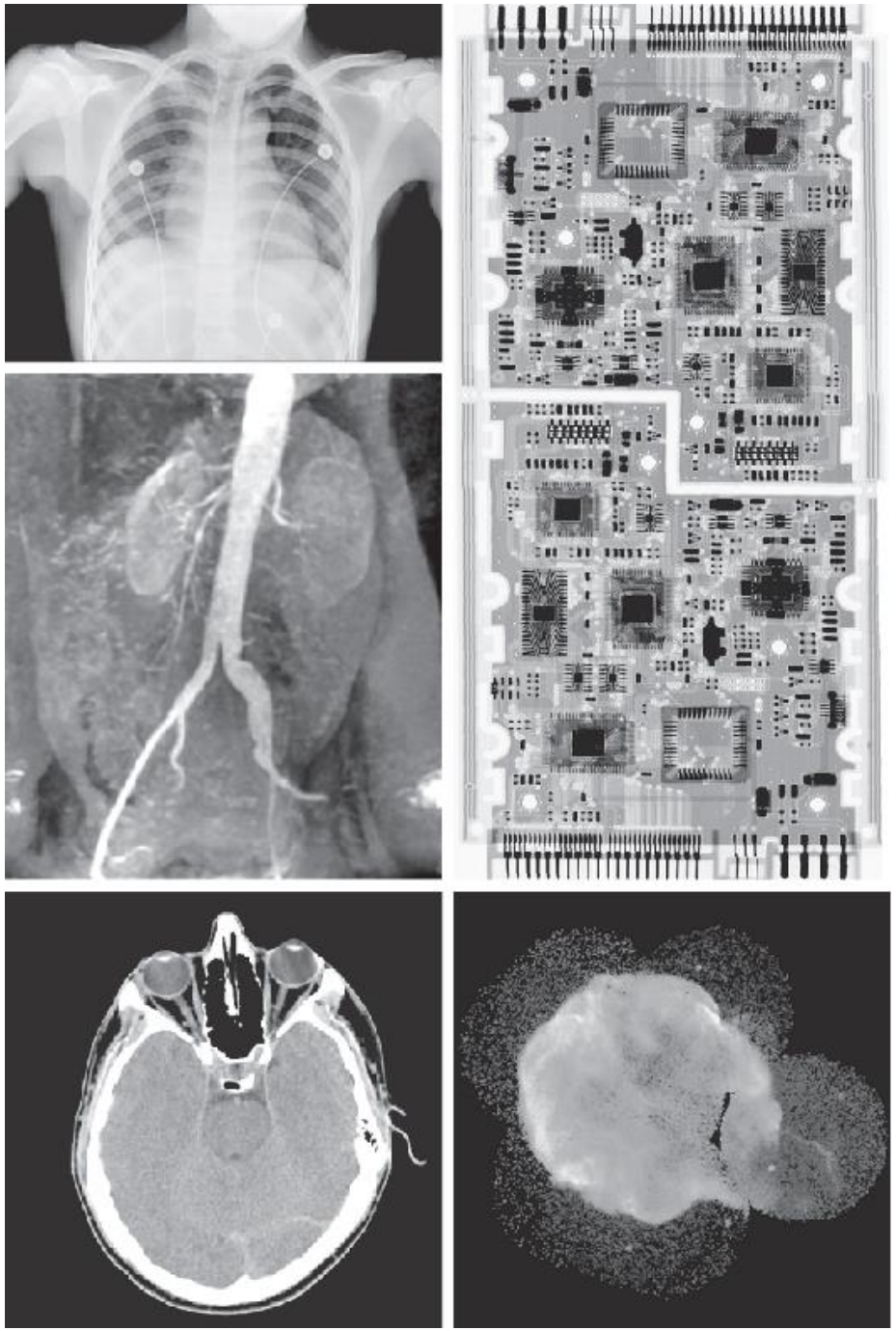

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FIGURE 1.7 Examples of X-ray imaging. (a) Chest X-ray. (b) Aortic angiogram. (c) Head CT. (d) Circuit boards. (e) Cygnus Loop. (Images courtesy of (a) and (c) Dr. David R. Pickens, Dept. of Radiology & Radiological Sciences, Vanderbilt University Medical Center; (b) Dr. Thomas R. Gest, Division of Anatomical Sciences, Univ. of Michigan Medical School; (d) Mr. Joseph E. Pascente, Lixi, Inc.; and (e) NASA.)

Imaging in Ultraviolet Band

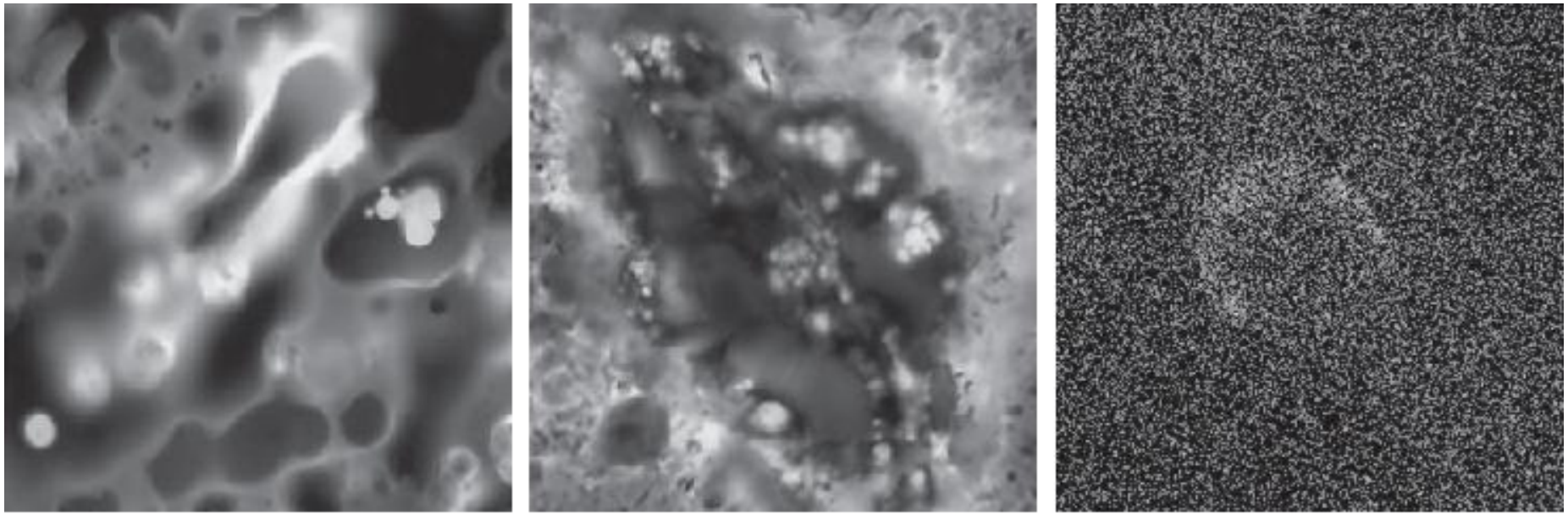

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FIGURE 1.8 Examples of ultraviolet imaging. (a) Normal corn. (b) Corn infected by smut. (c) Cygnus Loop. (Images (a) and (b) courtesy of Dr. Michael W. Davidson, Florida State University, (c) NASA.)

- X-ray Ultraviolet light is used in fluorescence microscopy
- Fluorescence: a phenomena in which a material called fluorescent emits visible light when ultraviolet light is directed at it
- Ultraviolet light is invisible, when directed on fluorescent material (photon on electron, energy level transition) visible light is emitted which is observed. High contrast in image w.r.t fluorescent to non-fluorescent region

Imaging in Visible Band – Light Microscopy

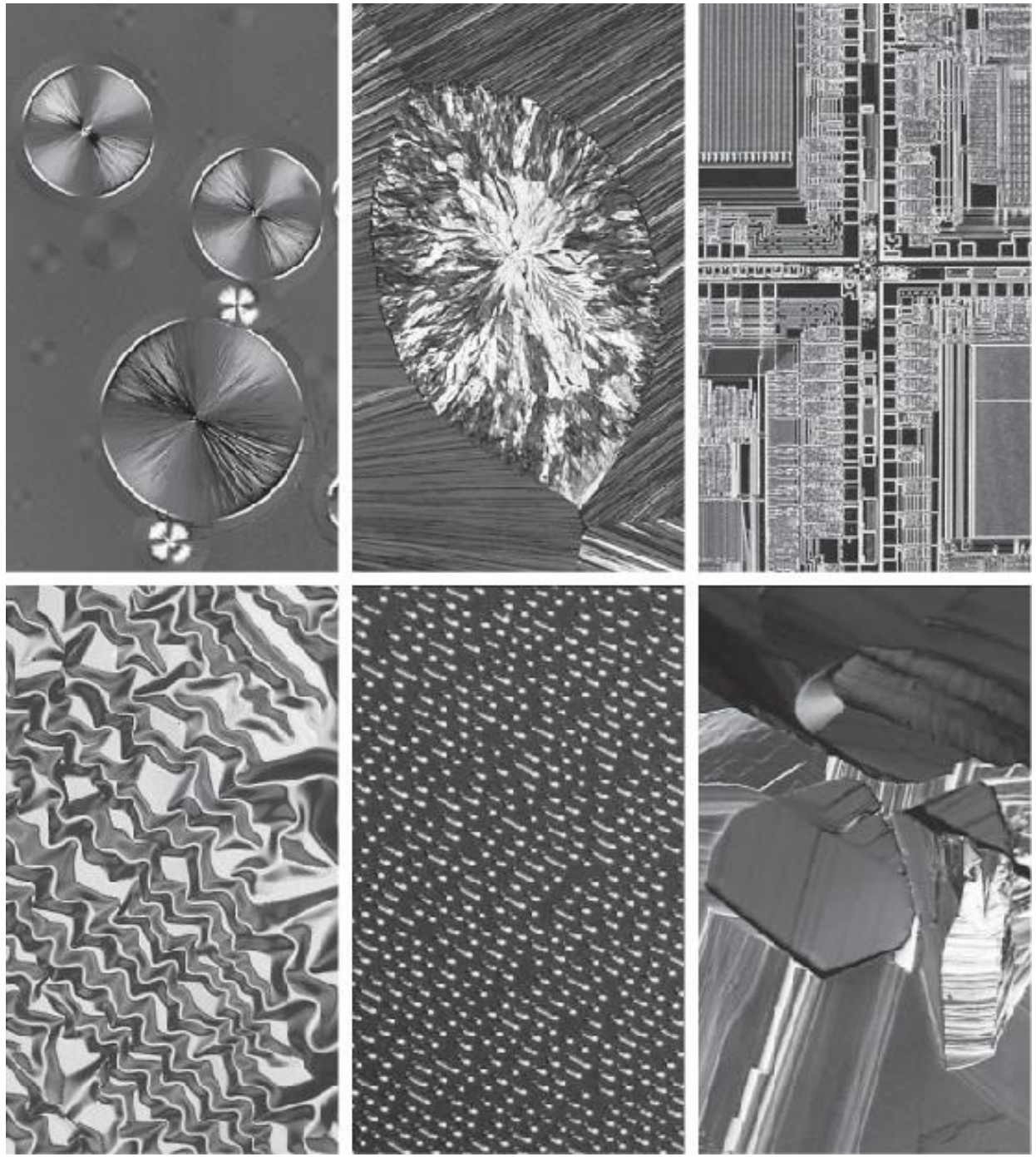

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FIGURE 1.9 Examples of light microscopy images. (a) Taxol (anticancer agent), magnified 250 ×. (b) Cholesterol—40 ×. (c) Microprocessor—60 ×. (d) Nickel oxide thin film—600 ×. (e) Surface of audio CD—1750 ×. (f) Organic superconductor— 450 ×. (Images courtesy of Dr. Michael W. Davidson, Florida State University.)

Imaging in Infrared/Visible Band

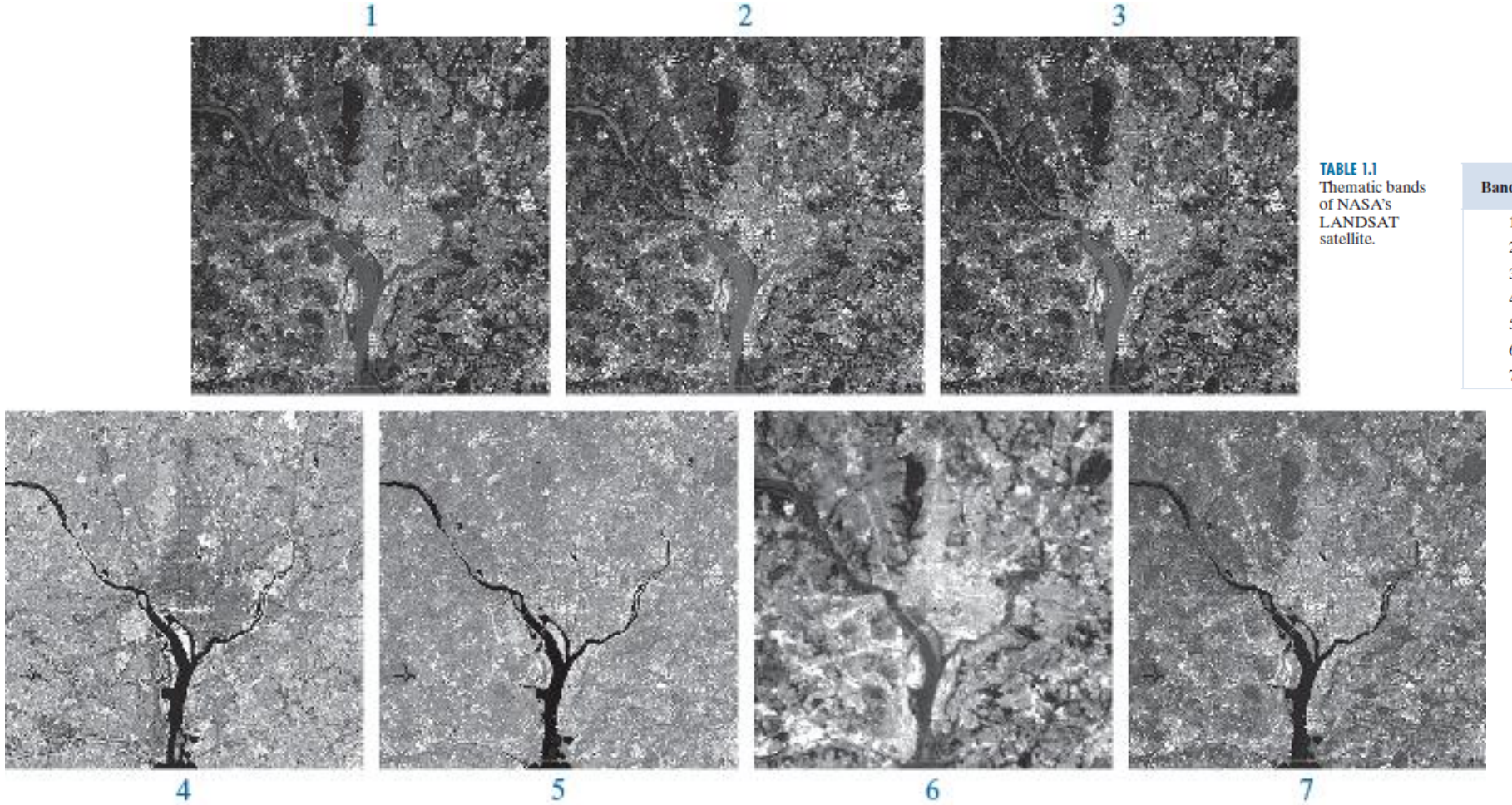

FIGURE 1.10 LANDSAT satellite images of the Washington, D.C. area. The numbers refer to the thematic bands in Table 1.1. (Images courtesy of NASA.)

TABLE 1.1 Thematic bands of NASA's LANDSAT satellite.

Band No.	Name	Wavelength (μm)	Characteristics and Uses
1	Visible blue	0.45–0.52	Maximum water penetration
2	Visible green	0.53–0.61	Measures plant vigor
3	Visible red	0.63–0.69	Vegetation discrimination
4	Near infrared	0.78–0.90	Biomass and shoreline mapping
5	Middle infrared	1.55–1.75	Moisture content: soil/vegetation
6	Thermal infrared	10.4–12.5	Soil moisture; thermal mapping
7	Short-wave infrared	2.09–2.35	Mineral mapping

- Remote sensing: to obtain images of the earth from space for purposes of monitoring environmental conditions
- Usually a scene is imaged in several bands

Imaging in Infrared Band

FIGURE 1.11 Satellite image of Hurricane Katrina taken on August 29, 2005. (Courtesy of NOAA.)

Infrared Imaging

- Images captured by infrared imaging system mounted on NOAA DMSP (Defense Military Satellite Program) satellite, provides a global inventory of human settlements

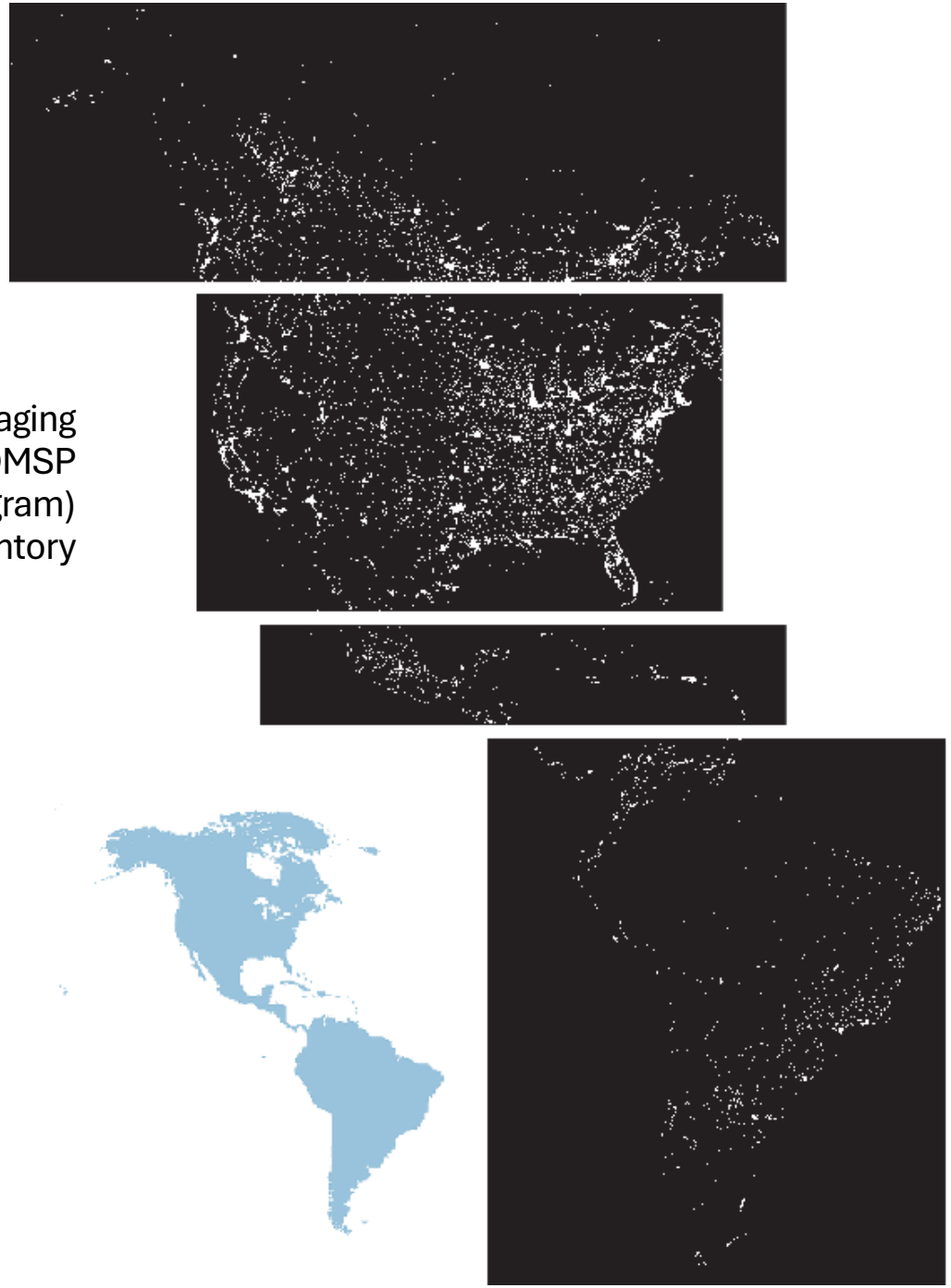

FIGURE 1.12 Infrared satellite images of the Americas. The small shaded map is provided for reference. (Courtesy of NOAA.)

Imaging in Infrared Band

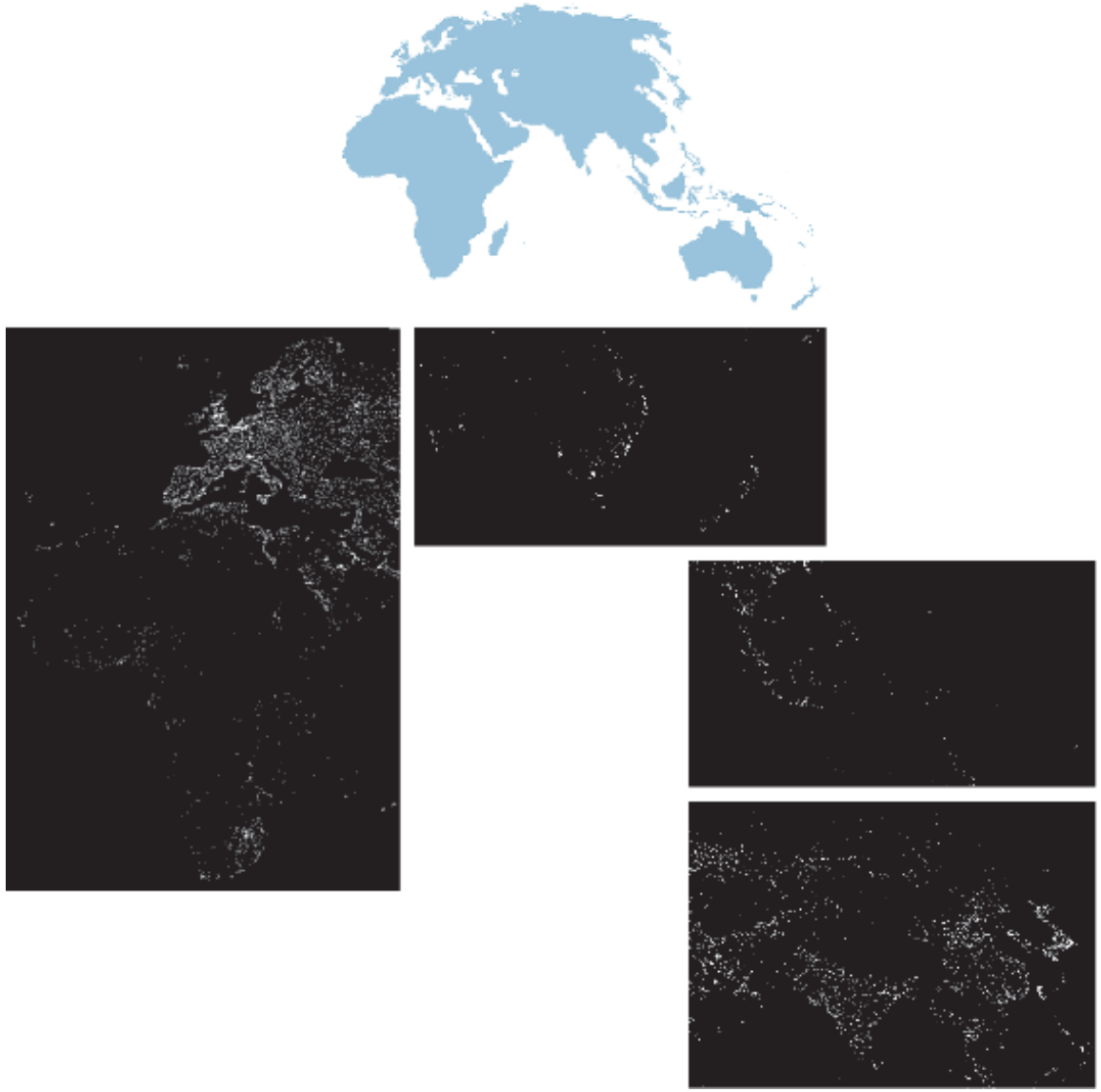

FIGURE 1.13 Infrared satellite images of the remaining populated parts of the world. The small shaded map is provided for reference. (Courtesy of NOAA.)

Imaging in Visible Band

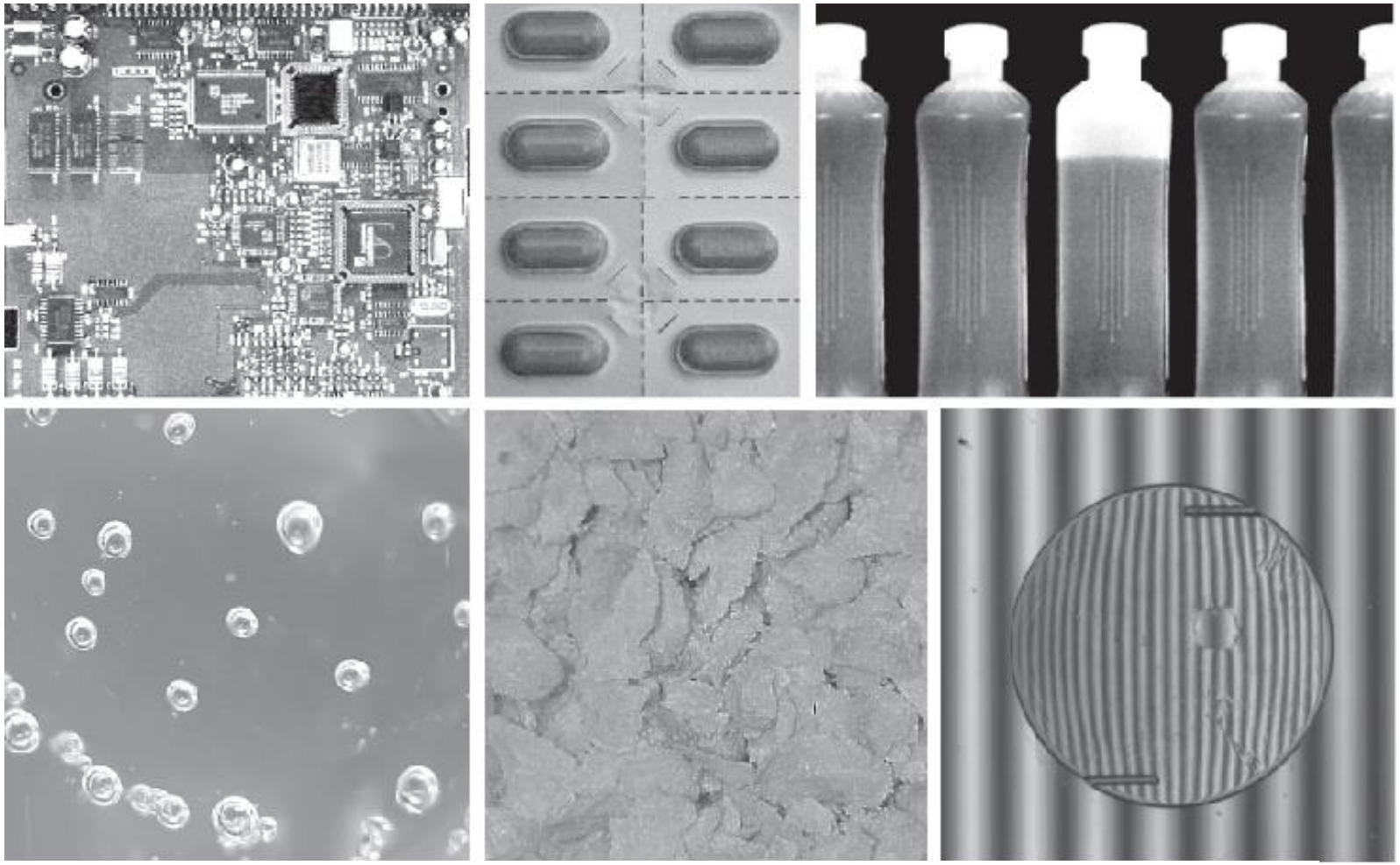

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FIGURE 1.14 Some examples of manufactured goods checked using digital image processing. (a) Circuit board controller. (b) Packaged pills. (c) Bottles. (d) Air bubbles in a clear plastic product. (e) Cereal. (f) Image of intraocular implant. (Figure (f) courtesy of Mr. Pete Sites, Perceptics Corporation.)

Imaging in Visible Band

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FIGURE 1.15 Some additional examples of imaging in the visible spectrum. (a) Thumb print. (b) Paper currency. (c) and (d) Automated license plate reading. (Figure (a) courtesy of the National Institute of Standards and Technology. Figures (c) and (d) courtesy of Dr. Juan Herrera, Perceptics Corporation.)

Imaging in Microwave Band

- The dominant application of imaging in the microwave band is RADAR where radar works like a flash camera (transmit microwave pulses and receive energy)
- Unique feature: able to collect data over virtually any region at any time regardless of weather or lighting condition

FIGURE 1.16 Spaceborne radar image of mountainous region in southeast Tibet. (Courtesy of NASA.)

Imaging in Radio Wave Band

- Magnetic Resonance Imaging (MRI): The radio waves pass through a patient' body in short pulses, which is kept in a high magnetic field.
- Each pulse causes a responding pulse emitted by the patient's tissue.
- The signal origin and strength are determined by a computer which produces a two dimensional image of a section of the patient.

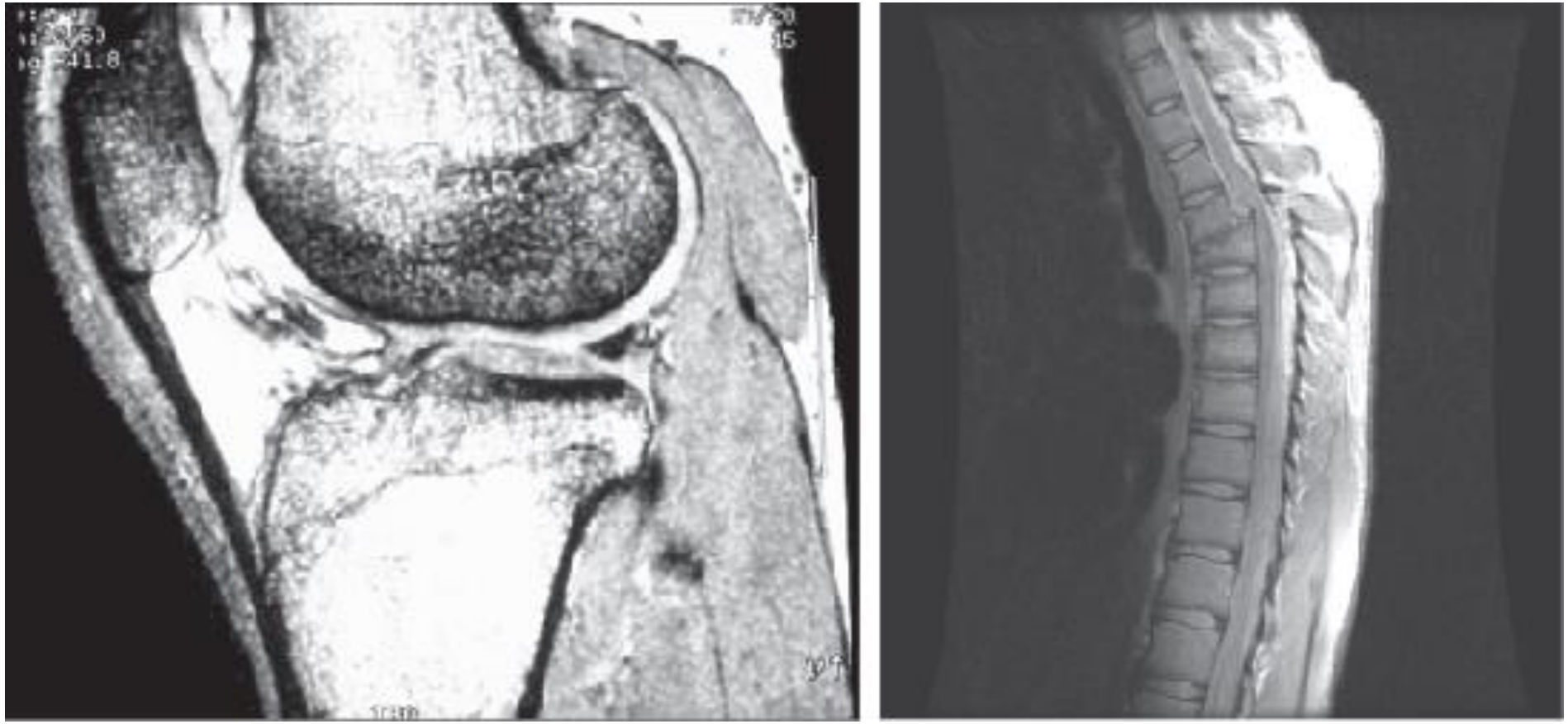

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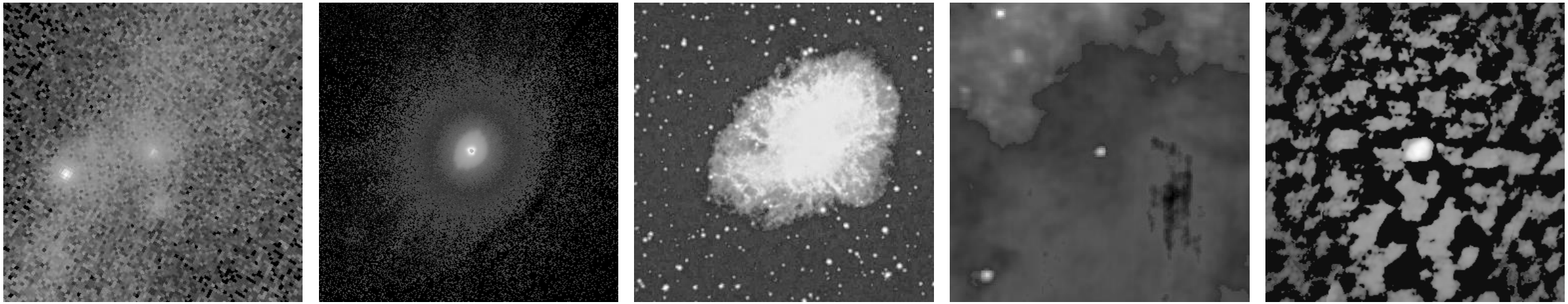
FIGURE 1.17 MRI images of a human (a) knee, and (b) spine. (Figure (a) courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School, and (b) courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

Crab Pulsar (a neutron star emitting energy in multiple EM spectrum bands)



Gamma X-ray Optical Infrared Radio

FIGURE 1.18 Images of the Crab Pulsar (in the center of each image) covering the electromagnetic spectrum. (Courtesy of NASA.)

Imaging using Sound Waves

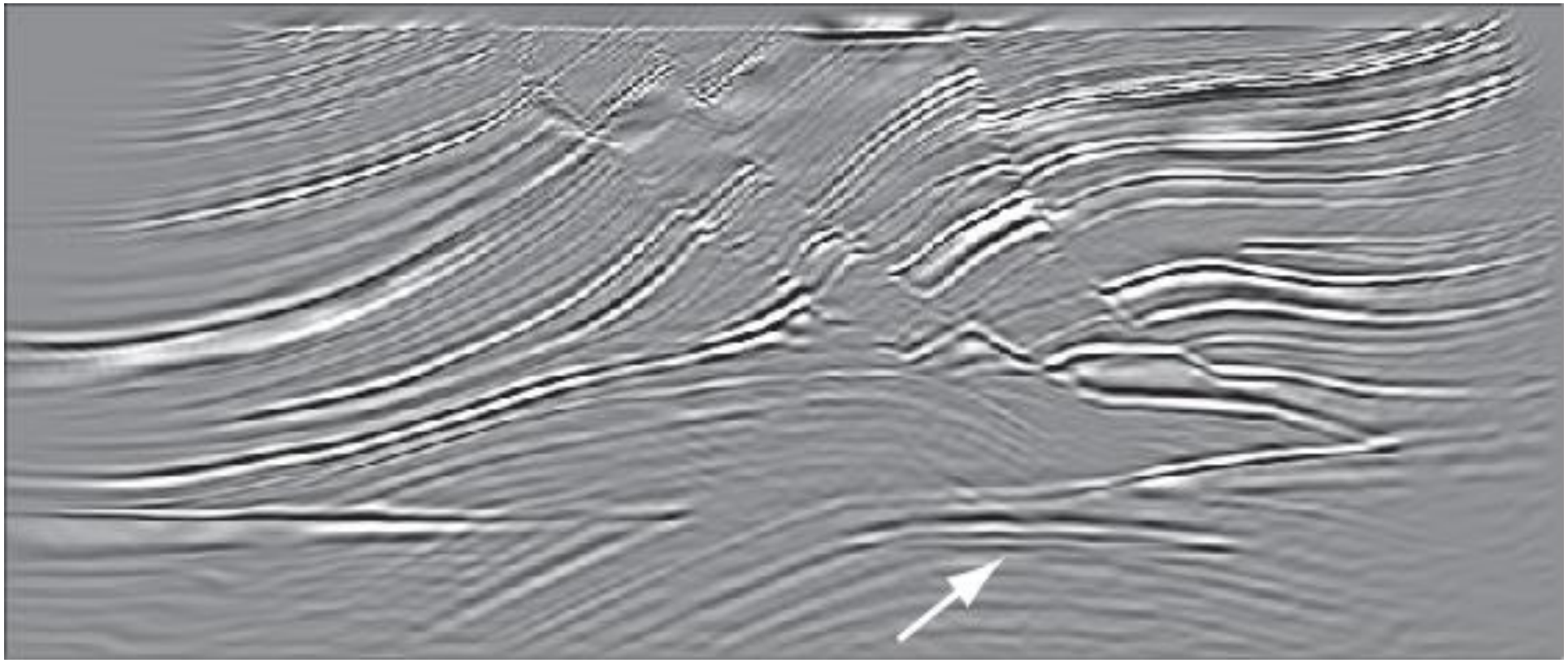


FIGURE 1.19 Cross-sectional image of a seismic model. The arrow points to a hydrocarbon (oil and/or gas) trap. (Courtesy of Dr. Curtis Ober, Sandia National Laboratories.)

Ultrasound Imaging

1. Ultrasound system transmits high-frequency (1 to 5 MHz) sound pulses into the body
2. The sound waves travel into the body and hit a boundary between tissues (e.g., soft tissue and bone). Some of the sound waves are reflected back to the probe, while some travel on further until they reach another boundary
3. The reflected waves are picked by the probe and relayed to a computer
4. The computer calculates the distance from the probe to the tissue using the speed of sound in tissue(1540 m/s)
5. The system displays the distance and intensities of the echoes on the screen, forming a two-dimensional image

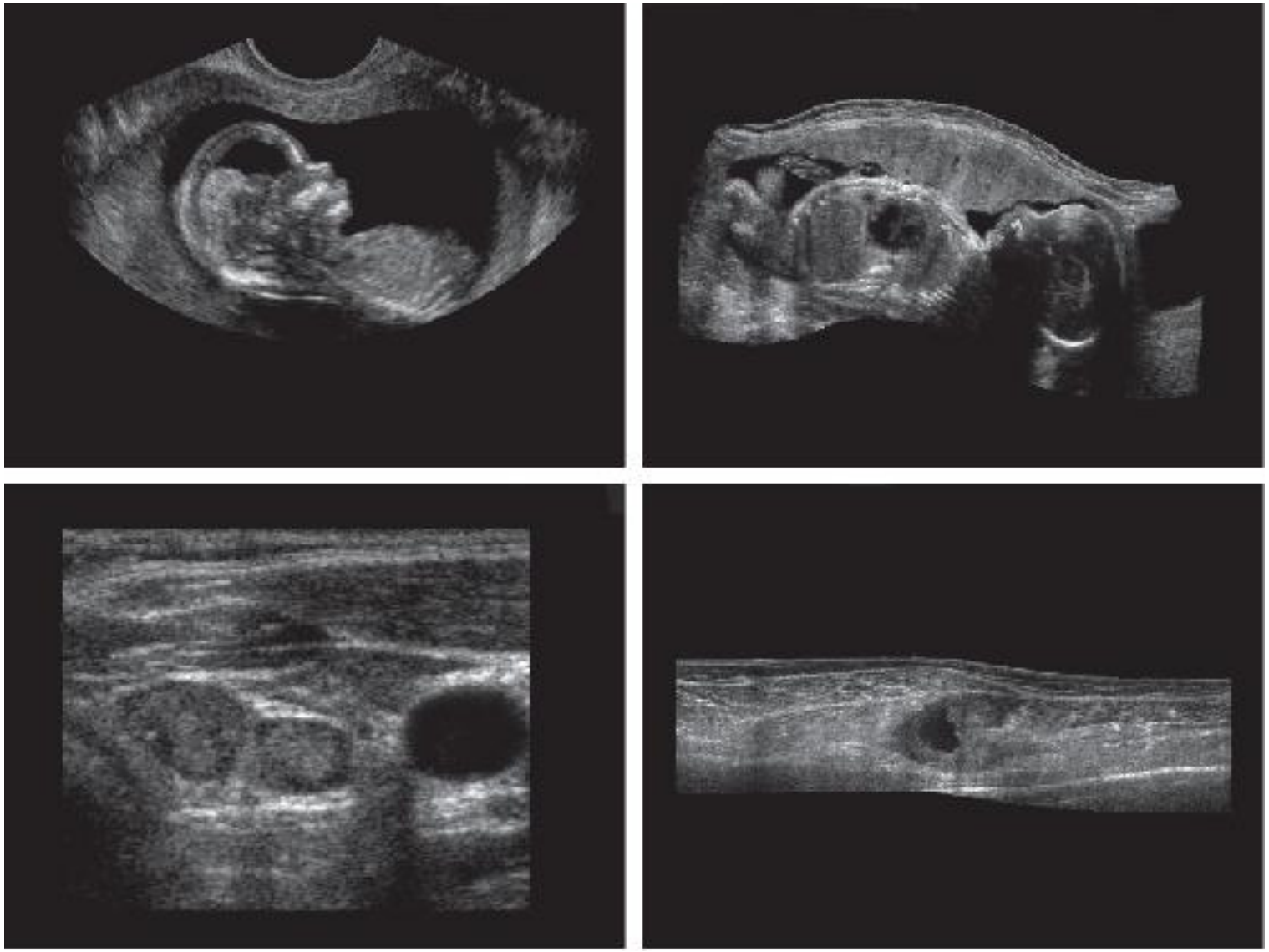


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FIGURE 1.20 Examples of ultrasound imaging. (a) A fetus. (b) Another view of the fetus. (c) Thyroids. (d) Muscle layers showing lesion. (Courtesy of Siemens Medical Systems, Inc., Ultrasound Group.)

Electron Microscopy

- TEM (Transmission Electron Microscope)
- SEM (Scanning Electron Microscope)
- Beam of electron pass though the slide projected on phosphor screen. Electrons interact with phosphor screen and produce light.
- Light Microscope: magnification 1000x
- Electron Microscope: magnification up to 10,000x and more

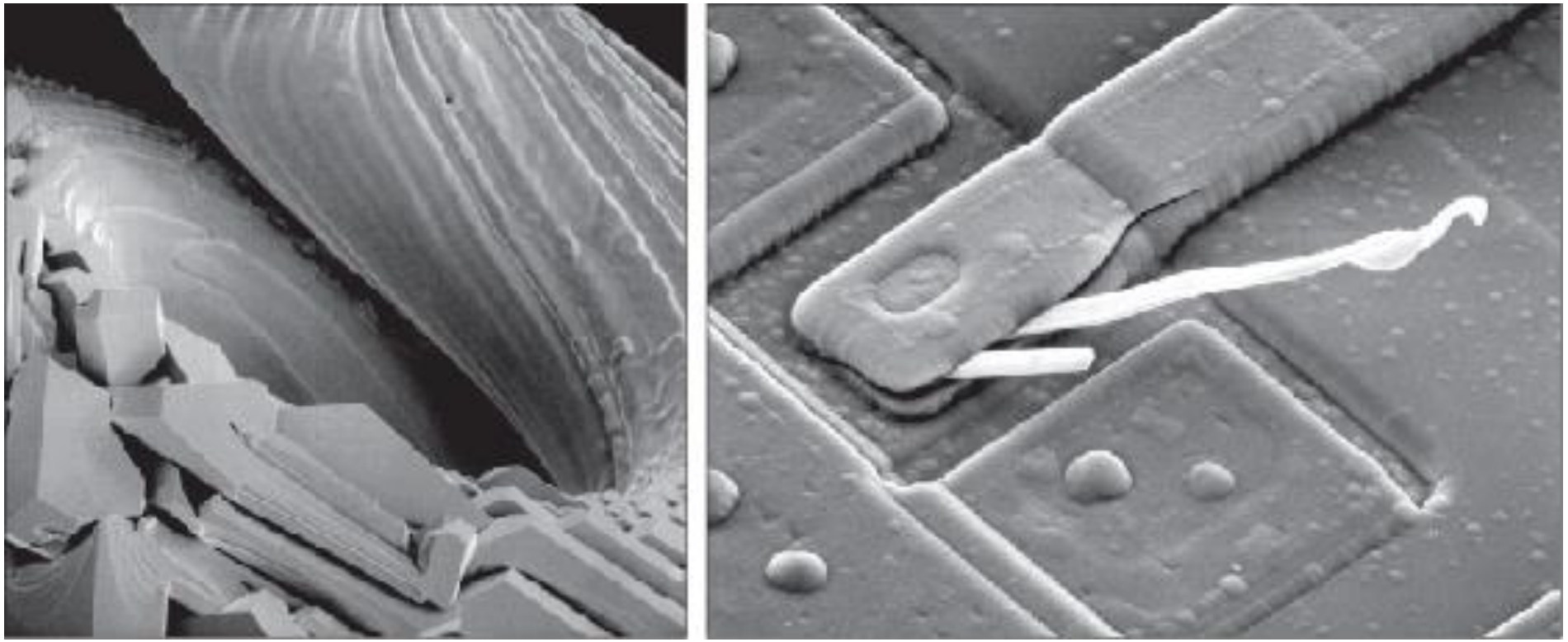

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FIGURE 1.21 (a) 250 × SEM image of a tungsten filament following thermal failure (note the shattered pieces on the lower left). (b) 2500 × SEM image of a damaged integrated circuit. The white fibers are oxides resulting from thermal destruction. (Figure (a) courtesy of Mr. Michael Shaffer, Department of Geological Sciences, University of Oregon, Eugene; (b) courtesy of Dr. J. M. Hudak, McMaster University, Hamilton, Ontario, Canada.)

Computer Generated Images

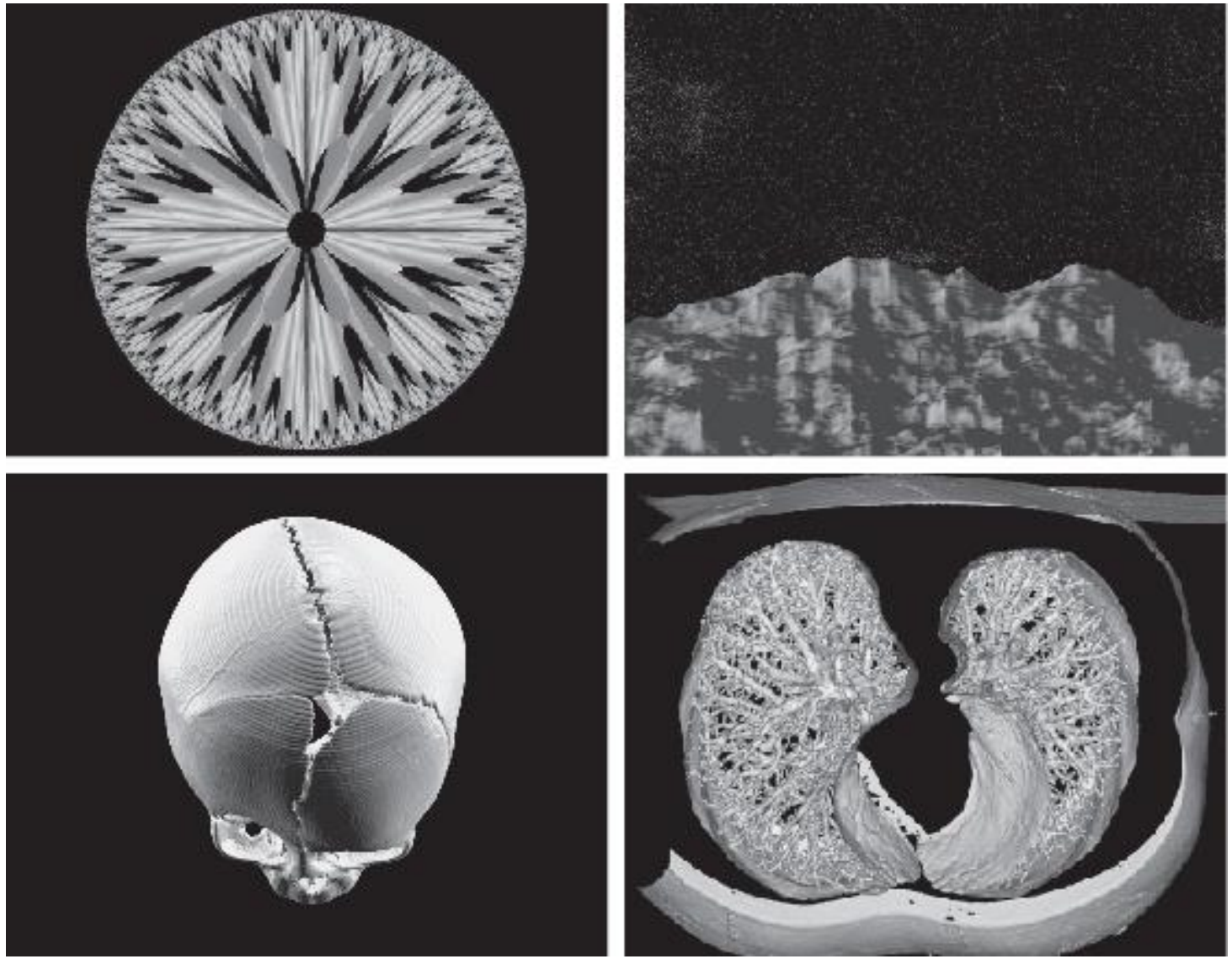

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FIGURE 1.22 (a) and (b) Fractal images. (c) and (d) Images generated from 3-D computer models of the objects shown. (Figures (a) and (b) courtesy of Ms. Melissa D. Binde, Swarthmore College; (c) and (d) courtesy of NASA.)

Fundamental Steps in Digital Image Processing

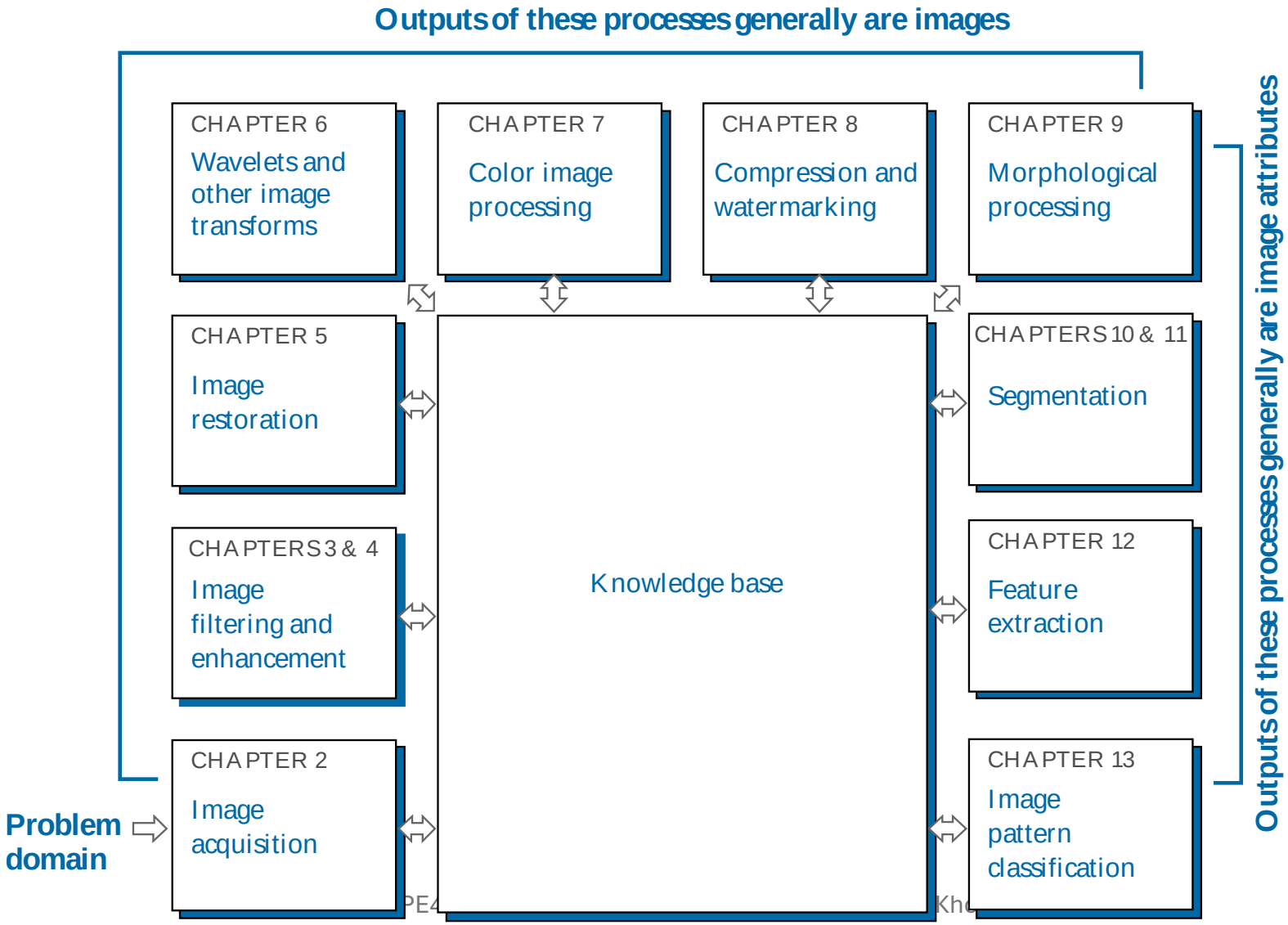

FIGURE 1.23 Fundamental steps in digital image processing. The chapter(s) indicated in the boxes is where the material described in the box is discussed.

Components of a Digital Image Processing System

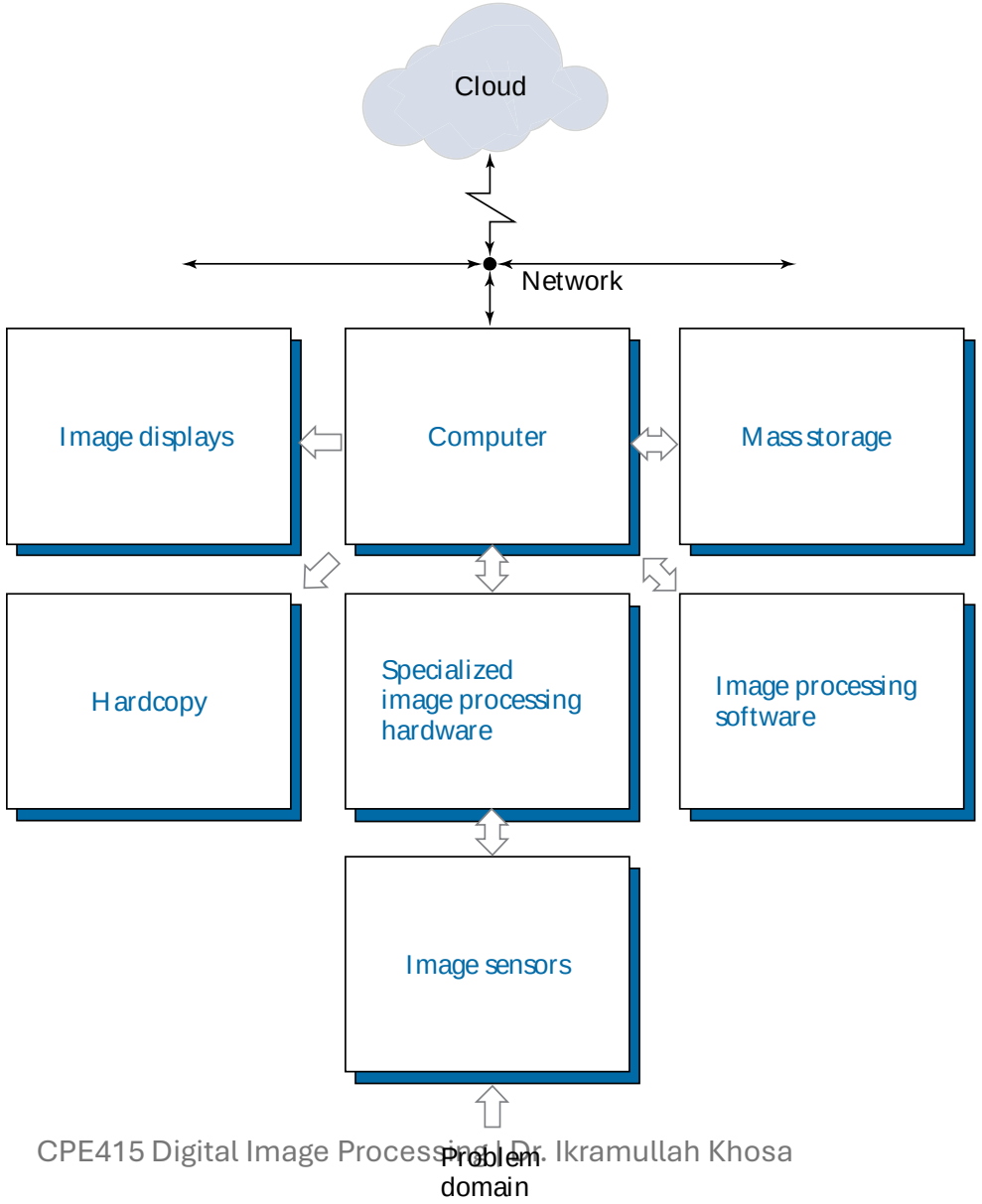

FIGURE 1.24 Components of a general-purpose image processing system.